THE MASTERMIND COACHING CLASSES

Revision Notes

Class 7 · Science

NCERT Curiosity — Chapters 7 to 11 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
7	Measurement of Time and Motion	Clocks, pendulum, time period, speed, uniform motion
8	Life Processes in Animals	Digestion, absorption, breathing, respiration, circulation
9	Life Processes in Plants	Photosynthesis, stomata, xylem and phloem, plant respiration
10	Light: Shadows and Reflections	Straight-line travel, shadows, plane mirror, pinhole camera
11	Earth, Moon, and the Sun	Rotation, revolution, seasons, solar and lunar eclipses

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CHAPTER 07

Measurement of Time and Motion

Prerna, a 100 m sprinter, wonders how time was measured before clocks, and how a winner is picked when two runners finish together. The chapter traces timekeeping from sundials and water clocks to pendulum, quartz and atomic clocks, studies the simple pendulum and its time period, and then introduces speed, its formula and units, average speed, and uniform and non-uniform linear motion.

Big Questions

- How did people measure time without clocks?
- Why does a pendulum keep time?
- How do we decide who is faster?
- What is the difference between uniform and non-uniform motion?

8.1 Measurement of Time

Early humans noticed natural events repeating at regular intervals: rising and setting of the Sun, phases of the Moon, changing seasons. These cycles gave **calendars** (a day = one cycle of sunrise to sunrise) and later devices to measure shorter intervals within a day.

Old time-measuring device	How it works
Sundial	Time read from the changing position of the shadow of an object cast by the Sun
Water clock (outflow type)	Water flows out of a marked vessel; level shows time
Water clock (floating bowl type)	A bowl with a fine hole floats, fills in a fixed time and sinks; then lifted and refloated
Hourglass	Sand flows from one bulb to another in a fixed time
Candle clock	A marked candle burns down at a steady rate

- **Samrat Yantra, Jantar Mantar, Jaipur:** the world's largest stone sundial, built about 300 years ago, 27 m high; its shadow moves about 1 mm per second on a scale reading intervals as short as 2 s. It shows local solar time, so a correction is needed to get Indian Standard Time. Jantar Mantar is a UNESCO World Heritage site.
- **Make a simple water clock (Activity 8.1):** cut a plastic bottle in two, make a small hole in the cap, invert the top over the bottom, fill with coloured water and mark the level after every minute using a watch. Each mark crossed later equals one minute.

Ancient India: shadow-based time measurement is mentioned in Kautilya's *Arthasastra*; Varahamihira (about 530 CE) gave an accurate expression for time from the shadow of a vertical stick. Outflow water clocks lost accuracy as the flow slowed with falling water level, leading to the sinking-bowl **Ghatika-yantra** (first mentioned by Aryabhata). Its hole was made so that the bowl filled and sank in **24 minutes = 1 ghatika (ghati)**, so a 24-hour day had **60 ghatis**. Each sinking was announced by drum, conch or gong. It continued at religious places even after pendulum clocks arrived in the late 19th century.

As people travelled long distances, accurate time became vital. Mechanical clocks driven by weights, gears and springs appeared from the 14th century. The **pendulum clock** (invented 1656, patented 1657 by

Christiaan Huygens) was a major breakthrough.

> NOTE
> **Know a scientist:** Galileo (1564–1642), watching a swinging lamp in a church and timing it with his pulse, found each swing took the same time. He concluded that the time for one oscillation is the same for a pendulum of a given length.

▸ 8.1.1 The simple pendulum

- A **simple pendulum** is a small metal ball (the **bob**) hung from a rigid support by a long thread.
- At rest the bob is at the **mean position O**. If moved slightly to one side (extreme position A) and released it oscillates to the other extreme B and back. This motion is **periodic**: it repeats after a fixed time.
- **One oscillation:** O → A → O → B → O, or A → B → A.

> DEFINITION
> The **time period** of a pendulum is the time taken to complete **one oscillation**.

Set up (Activity 8.2)
Tie a bob to a string about 150 cm long and fix the other end to a rigid support so that the length between support and bob is about 100 cm. Let it come to rest.

Release gently
Move the bob slightly to one side and release it without pushing, keeping the string taut.

Time 10 oscillations
Measure the time for 10 oscillations with a watch. Repeat 3–4 times.

Calculate
Time period = Time for 10 oscillations ÷ 10.

Conclude
The time period is nearly the same every time.

> FORMULA
> Time period = (Time taken for n oscillations) ÷ n

Think like a scientist: repeat with different lengths and with bobs of different masses. Result: **the time period of a simple pendulum depends on its length but not on the mass of the bob.** At a given place, pendulums of the same length have the same time period, which is why pendulums are used for timekeeping.

- Every clock depends on some process repeating regularly to mark equal intervals.
- Modern clocks use tiny, rapid vibrations of a **quartz crystal** (quartz clocks) or of specific **atoms** (atomic clocks). Huygens' clocks gained or lost about 10 s a day; atomic clocks lose about 1 s in millions of years.

▸ 8.1.2 SI unit of time

Quantity	Unit	Symbol	Relation
Time	second (SI)	s	Base SI unit
Time	minute	min	60 s = 1 min
Time	hour	h	60 min = 1 h

IMPORTANT

Writing rules: unit names (second, minute, hour) are in lowercase; symbols (s, min, h) are lowercase and singular (not 'hrs' or 'sec'); no full stop after a symbol (except at the end of a sentence); leave a space between number and unit, e.g. 5 s.

- A normal wall clock can measure down to **1 second**.
- **Science and Society:** sports timers record 1/100 or 1/1000 s (millisecond); ECG machines record millisecond variations in heartbeat; digital audio is sampled thousands of times a second; computers and phones work in microseconds (one-millionth of a second).

8.2 Slow or Fast

In a race the runner who covers more distance in the same time is faster. To compare movements over different distances we need **speed**.

8.3 Speed

DEFINITION

Speed is the distance covered by an object in unit time (one second, minute or hour).

FORMULA

Speed = Total distance covered ÷ Total time taken. **Distance = Speed × Time. Time = Distance ÷ Speed.**

- **SI unit:** metre per second (**m/s**). Other common unit: kilometre per hour (**km/h**).
- **Conversion:** 1 km/h = 5/18 m/s; 1 m/s = 3.6 km/h (since 1 km = 1000 m, 1 h = 3600 s).
- The speed found by total distance ÷ total time is the **average speed**; the textbook uses 'speed' to mean average speed.
- **Speedometer** shows a vehicle's speed in km/h; **odometer** shows the distance travelled in km.

▸ Worked examples

Problem	Solution
Swati cycles 3.6 km to school in 15 min. Find speed in m/s.	3.6 km = 3600 m; 15 min = 900 s; speed = 3600 ÷ 900 = **4 m/s**
A bus at 50 km/h travels for 2 h. Distance?	Distance = 50 × 2 = **100 km**
A train at 90 km/h must cover 360 km. Time?	Time = 360 ÷ 90 = **4 h**
A car goes 150 m in 10 s. Speed in km/h?	150 ÷ 10 = 15 m/s = 15 × 3.6 = **54 km/h**
A train goes 180 km in 3 h. Speed? Distance in 4 h?	Speed = 60 km/h = 60 × 5/18 = about 16.7 m/s; in 4 h = 60 × 4 = **240 km**
Runner A: 400 m in 50 s; Runner B: 400 m in 45 s. Who is faster?	A = 8 m/s; B = about 8.89 m/s. **B is faster** by about 0.89 m/s
A car covers 60, 70 and 50 km in three successive hours. Average speed?	Total 180 km in 3 h = **60 km/h**. Not uniform, since distances in equal times differ

> **NOTE**
> Compare speeds in the same units, e.g. a horse at 18 m/s versus a train at 72 km/h = 20 m/s, so the train is faster.

Activity 8.4: pick a train from a railway timetable, find the distance to the next station and the time taken between the stations, and compute its speed. The train with the highest speed is the fastest.

8.4 Uniform and Non-uniform Linear Motion

Motion along a straight line is **linear motion**. A train going from station A to D starts slowly, speeds up, then runs steadily from B to C, then slows to a halt at D.

	Uniform linear motion	Non-uniform linear motion
Speed	Constant (unchanging)	Keeps changing
Distances in equal time intervals	Equal	Unequal
Example	Train between B and C; car on an empty highway	Train between A and B and C and D; car in city traffic; most everyday motion

From the table of trains X and Y over 10-minute intervals, X covers equal distances each interval (uniform) whereas Y does not (non-uniform). Uniform motion is an **idealisation**: in everyday life speed changes, so **average speed** is used.

Likely exam questions

Likely question	Model answer
Name four ancient devices for measuring time.	Sundial, water clock, hourglass and candle clock.
What is the Ghatika-yantra and what unit did it give?	Sinking-bowl water clock that sank in 24 minutes; one ghatika = 24 min, 60 ghatis in a day.
What is a simple pendulum? Define time period and one oscillation.	A bob on a thread from a rigid support; time period is the time for one oscillation (e.g. O to A to O to B to O).
How do you find the time period experimentally?	Time 10 oscillations and divide by 10; repeat a few times.
On what does the time period of a simple pendulum depend?	Only on its length (at a place), not on the bob's mass.
Who invented the pendulum clock and who inspired him?	Christiaan Huygens (1656), inspired by Galileo's study of pendulums.
Name modern clocks. Why are they accurate?	Quartz and atomic clocks; they use very rapid regular vibrations of quartz or atoms.
Write the SI unit of time and the writing rules for units.	Second (s); units lowercase, symbols singular and without full stop, space between number and unit; 'sec' and 'hrs' are wrong.
Define speed and give its SI unit.	Distance covered in unit time; metre per second (m/s).
Convert 72 km/h into m/s.	72 × 5/18 = 20 m/s.

Likely question	Model answer
What are speedometer and odometer?	Instruments showing a vehicle's speed (km/h) and distance travelled (km).
Differentiate uniform and non-uniform motion.	Uniform: constant speed, equal distances in equal times; non-uniform: speed changes, unequal distances in equal times.
Which is more common in daily life and why do we use average speed?	Non-uniform motion; speed keeps changing, so total distance ÷ total time is used.
A car travels 2 km: first 500 m at 10 m/s, next 500 m at 5 m/s. Speed for the rest to finish in 200 s?	Times: 50 s and 100 s; remaining 1000 m in 50 s = 20 m/s. Average speed = 2000 ÷ 200 = 10 m/s.

Exploratory projects

- Build a floating-bowl water clock; vary bowl and hole size to get a 24-minute sinking time.
- Measure the time period of playground swings with children of different weights and swings of different lengths (a swing is a pendulum).
- Compare the speeds of Olympic 100 m, 200 m and 400 m winners.
- Find sources of variation in pendulum readings and reduce them.

★ KEY TAKEAWAYS

- Early time-keeping used repeating natural cycles and devices: **sundial, water clock, hourglass, candle clock**.
- A pendulum's **time period** depends on its length, not on the bob's mass; this led to **pendulum clocks**; modern clocks use quartz and atoms.
- SI unit of time: **second (s)**; 60 s = 1 min; 60 min = 1 h.
- **Speed = distance ÷ time** (m/s or km/h); 1 m/s = 3.6 km/h; **distance = speed × time**; **time = distance ÷ speed**.
- Total distance ÷ total time is **average speed**.
- **Uniform linear motion:** constant speed, equal distances in equal times; otherwise **non-uniform**.

CHAPTER 08

Life Processes in Animals

Animals take in food, break it down, get energy from it, breathe and transport materials around the body. These essential activities, nutrition, circulation, respiration, excretion and reproduction, are the life processes. This chapter studies in detail how humans digest food (nutrition) and breathe and respire, and then compares other animals such as ruminants, birds, fish, frogs and earthworms.

Big Questions

- Why does chapati taste sweet if chewed long?
- Where is digested food absorbed?
- How is breathing different from respiration?
- How do fish, frogs and earthworms breathe?

9.0 Introduction

Life processes (nutrition, circulation, respiration, excretion, reproduction) are needed for survival. Animals eat many kinds of food: bees and sunbirds suck nectar, infants drink milk, pythons swallow prey whole, some aquatic animals filter tiny food particles. Food gives **energy** for all life processes.

Food contains complex components (**carbohydrates, proteins, fats**) that must be broken into simpler forms. This happens in the **alimentary canal**, a long tube from the **mouth to the anus**. Digestive juices secreted along it break food down; the simple forms are absorbed and carried to the body.

9.1 Nutrition in Animals

▸ 9.1.1 Digestion in human beings

DEFINITION

Digestion is the breakdown of complex food components into simpler forms in the body. **Mechanical digestion** is the initial breakdown of food into fine pieces by teeth through crushing and chewing.

▸ Mouth cavity

- Teeth do mechanical digestion; the tongue mixes food with **saliva**; saliva moistens food, making it soft and easy to swallow.
- Saliva contains a digestive juice that breaks **starch into sugar**. This is why chapati or rice tastes sweet if chewed for 30–60 s. Food is partly digested in the mouth.
- **Oral hygiene:** brush teeth and clean the tongue twice a day, rinse after every meal to prevent decay and bad smell.

Set up (Activity 9.1)
Label test tubes A and B. Put a teaspoon of boiled rice in A and a teaspoon of rice chewed for 30–60 s in B. Add 3–4 mL water to each.

Add iodine
Add 3–4 drops of iodine solution to each and mix.

> **Observe**
> A turns **blue-black** (starch present). B shows no change or only light blue-black (starch has been broken into sugars by saliva).

Iodine gives a blue-black colour with starch. Chewing for longer should reduce the colour further.

▸ Food pipe (oesophagus)

A long flexible tube from mouth to stomach. Its walls **contract and relax in wave-like motion** to push food down. Such movement occurs along the entire alimentary canal.

▸ Stomach

- Walls contract and relax to **churn** food, which mixes with secretions from the stomach lining: **digestive juice, acid and mucus**.
- Digestive juice breaks down **proteins**; **acid** helps in this and kills many harmful bacteria; **mucus** protects the stomach lining from the acid.
- Food becomes a **semi-liquid mass**, partially digested.
- **Fascinating fact:** in 1822 Alexis St. Martin was shot in the stomach and his wound left a permanent opening, which let Dr William Beaumont observe digestion directly and study how foods and emotions affect it.

▸ Small intestine

- About **6 m long**, the longest part of the alimentary canal; called 'small' because it is narrow.
- Receives secretions from **three sources**: its own lining, the **liver** and the **pancreas**.
- **Liver:** makes **bile**, mildly basic; it neutralises acid coming from the stomach and breaks fats into tiny droplets.
- **Pancreas:** makes **pancreatic juice**, basic, which neutralises acid and breaks down carbohydrates, proteins and fats.
- Juice from the small intestine wall further breaks down fats, proteins and partly digested carbohydrates.
- **Absorption:** digested nutrients pass into blood vessels in the intestinal walls. The inner lining has thousands of **finger-like projections (villi)** that increase surface area for absorption.
- Blood carries nutrients for energy, growth and repair.
- **Celiac disease:** the body reacts to **gluten** (a protein in wheat, barley, rye), damaging the intestinal lining. It is managed by avoiding gluten; millets (jowar, bajra, ragi) are naturally gluten-free.

▸ Large intestine

- About **1.5 m long** but **wider** than the small intestine.
- Absorbs **water and some salts** from the undigested food, making the waste semi-solid **stool**.
- Stool is stored in the **rectum** and expelled through the **anus**, a process called **egestion**.
- Fibre-rich foods (fruits, vegetables, whole grains) make stool easier to pass.
- Helpful bacteria in the large intestine break down undigested food (especially fibre) and make useful nutrients. Fermented foods (curd, buttermilk, shrikhand, kanji, pickles, gundruk, poita bhat) support gut health.
- The *Charaka Samhita* stresses easily digestible food and spices like ginger, black pepper and cumin; modern advice: eat on time, eat mindfully, avoid overeating.

Part of alimentary canal	Key function
Mouth	Chewing (mechanical), saliva acts on starch

Part of alimentary canal	Key function
Oesophagus	Moves food to stomach by wave-like contractions
Stomach	Churns food; acid, mucus, juice digest proteins; semi-liquid mass
Small intestine	Bile, pancreatic and intestinal juices complete digestion; **absorption** into blood
Large intestine	Absorbs water and salts; forms stool
Rectum and anus	Storage and egestion of stool
Liver, pancreas (associated organs)	Make bile and pancreatic juice

▸ 9.1.2 Do all animals digest food the same way?

- **Birds** have no teeth; they have a muscular **gizzard** that grinds food with the help of swallowed grit (small stones).
- **Ruminants** (cows, buffaloes, goats) partly chew grass and swallow it into the stomach (rumen), where it is partially digested. The food is brought back to the mouth and chewed thoroughly: **rumination**. A cow chews about 8 hours a day.
- Alimentary canals vary in structure and function to suit different foods.

9.2 Respiration in Animals

Some nutrients build and repair the body; others, like sugar, are broken down to release **energy**. The process of converting nutrients into usable energy is **respiration**.

▸ 9.2.1 Respiration in humans

> DEFINITION
> **Breathing** is the process of inhaling and exhaling air. The **respiratory system** carries out breathing and gas exchange.

Without food we can live for weeks and without water for a day or two, but without breathing only for a few minutes.

Path of air: nostrils → nasal passages (hair and mucus trap dust, so breathe through the nose) → windpipe → two branches into the two lungs → smaller branches ending in balloon-like **alveoli**. The lungs are protected by the **rib cage**; below them is the dome-shaped muscle, the **diaphragm**.

- **Science and Society:** small infectious particles can get past the filters; the SARS-CoV-2 virus of COVID-19 affected the respiratory system.

▸ Mechanism of breathing

	Inhalation (breathing in)	Exhalation (breathing out)
Ribs	Move up and outwards	Move down and inwards
Diaphragm	Moves **downwards**	Moves **upwards**
Chest space	Increases	Decreases
Air	Drawn into lungs	Pushed out of lungs

Model (Activity 9.2): a bottle with its base removed, a Y-tube with two balloons in the neck and a rubber sheet at the base. Pulling the sheet down inflates the balloons (inhalation); releasing it deflates them (exhalation). **Balloons = lungs; rubber sheet = diaphragm.**

- Breathing practices: Pranayama, Ladakh's Tummo breathing and deep breathing help respiratory health, relaxation and concentration.

▸ What do we breathe out? (Activity 9.3, teacher demonstration)

Equal fresh lime water in tubes A and B. Pass ordinary air with a pichkari into A, blow exhaled air into B with a straw. Only **B turns milky**, so exhaled air contains more carbon dioxide.

Gas	Inhaled air	Exhaled air
Oxygen	About 21%	About 16–17%
Carbon dioxide	About 0.04%	About 4–5%

Not all oxygen is used up in the lungs; some animals use a larger fraction.

▸ Gas exchange in alveoli

Alveoli have thin walls surrounded by fine blood vessels. **Oxygen** passes from alveoli into blood and is carried to all body parts; **carbon dioxide** from the body is carried by blood to the alveoli and released into the air to be breathed out.

▸ Respiration: using oxygen to release energy

> **FORMULA**
> Glucose + Oxygen → Carbon dioxide + Water + Energy

Breathing	Respiration
Taking in air and giving out air	Using oxygen to break down glucose and release energy
Physical process	**Chemical** process
Happens in the lungs/respiratory system	Happens inside body cells
Brings in oxygen, removes carbon dioxide	Gives energy for walking, running, playing, thinking

▸ Circulatory system

Made of the **heart, blood and blood vessels**. The heart pumps blood through the vessels to carry nutrients, oxygen and other substances to all parts and carry away waste.

> **IMPORTANT**
> **Smoking** damages lungs, causes persistent cough, frequent infections and lung cancer. **Passive smoking** (inhaling others' smoke) is especially dangerous for children, pregnant women and the elderly.

▸ 9.2.2 Breathing in other animals

Animal	Breathing organ
Birds, elephants, lions, cows, goats, lizards, snakes	Lungs (structure differs)

Animal	Breathing organ
Fish and most aquatic animals	**Gills**, rich in blood vessels; exchange gases dissolved in water
Frog tadpole	Gills
Adult frog	Lungs on land and **skin** in water
Earthworm	Moist **skin**

Likely exam questions

Likely question	Model answer
What are life processes?	Processes essential for survival: nutrition, circulation, respiration, excretion, reproduction.
What is the alimentary canal?	Long tube from mouth to anus in which food is digested and absorbed.
Why does chapati taste sweet on chewing?	Saliva breaks starch into sugar.
What does iodine test show in the saliva experiment?	Rice alone turns blue-black (starch); chewed rice shows little or no colour as saliva has digested starch.
What are the functions of acid, mucus and juice in the stomach?	Juice breaks proteins; acid helps and kills bacteria; mucus protects stomach lining.
Why is the small intestine called small, and what are its functions?	It is narrow although 6 m long; it completes digestion and absorbs nutrients through villi.
What do the liver and pancreas contribute?	Liver gives bile (neutralises acid, breaks fats); pancreas gives pancreatic juice (neutralises acid, digests carbohydrates, proteins, fats).
What happens in the large intestine?	Water and salts are absorbed and stool is formed and stored in the rectum before egestion.
What is rumination? Name some ruminants.	Bringing back partly digested food for thorough chewing; cow, buffalo, goat.
How do birds grind food without teeth?	Using the gizzard and swallowed grit.
Trace the path of air to the alveoli.	Nostrils, nasal passages, windpipe, bronchial branches in lungs, alveoli.
What happens to the ribs and diaphragm in inhalation and exhalation?	Inhalation: ribs up and out, diaphragm down. Exhalation: ribs down and in, diaphragm up.
What do the balloons and rubber sheet represent in the model?	Lungs and diaphragm.
Differentiate breathing and respiration.	Breathing is physical, exchange of air; respiration is chemical, releasing energy from glucose with oxygen.
Write the word equation for respiration.	Glucose + Oxygen gives carbon dioxide + water + energy.
How do fish, frogs and earthworms breathe?	Gills; lungs/skin (tadpoles gills); moist skin.

Exploratory projects

- Report on traditional oral hygiene practices.
- Make a 3-D clay model of the digestive system with labels.
- Find out what AQI is and how air quality affects farmers, factory workers and street vendors.
- Learn about box-breathing and its benefits; think about how birds cope with low oxygen at high altitude.

★ KEY TAKEAWAYS

- Digestion breaks food into simple forms in the **alimentary canal**: mouth → oesophagus → stomach → small intestine → large intestine → anus.
- **Saliva** digests starch; **stomach** juice digests protein; **bile, pancreatic and intestinal juices** complete digestion in the small intestine.
- **Absorption** occurs in the **small intestine**; the large intestine absorbs water and salts.
- Ruminants chew cud; birds use a gizzard.
- Breathing: ribs and **diaphragm** change chest volume; gas exchange occurs in **alveoli**.
- **Respiration:** glucose + oxygen → carbon dioxide + water + energy; breathing is physical, respiration chemical.
- The **circulatory system** (heart, blood, vessels) transports nutrients and oxygen.
- Fish use gills, frogs use skin and lungs, earthworms use moist skin.

CHAPTER 09

Life Processes in Plants

Animals eat food, but plants do not. This chapter asks how plants grow and how they make their own food. Through a series of experiments it shows that sunlight, water, chlorophyll and carbon dioxide are needed for photosynthesis, that oxygen is released, that stomata exchange gases, that xylem and phloem transport water, minerals and food, and that plants also respire.

Big Questions

- What do plants need in order to grow?
- Where and how do plants make food?
- How do water and food travel within a plant?
- Do plants respire like us?

10.1 How Do Plants Grow?

As a plant grows its height and stem thickness increase and new leaves and branches appear. Students suggest that water, sunlight or soil nutrients may matter. Activity 10.1 tests this.

Set up (Activity 10.1)
Take three identical pots with saplings of similar size (e.g. chilli or tomato). Label them A, B, C and count leaves at the start.

Vary conditions
Pot A: direct sunlight with water. Pot B: direct sunlight, no water. Pot C: dark, with water.

Observe two weeks
Note height, number of leaves and leaf colour.

Pot	Sunlight	Water	Result
A	Yes	Yes	Best growth
B	Yes	No	Plant may die
C	No	Yes	Poor growth, yellowish leaves

IMPORTANT

Conclusion: plants need both **sunlight and water** for growth.

Vrikshayurveda, an ancient Indian text, records observations on plant growth, soil, organic manure preparation and farming practices; it notes that trees do not bear flowers and fruits merely because they are planted.

10.2 How Do Plants Get Food for Their Growth?

Animals depend on plants directly or indirectly, but plants do not eat; they make their own food.

▸ 10.2.1 Leaves: the food factories

- Plants store food as **starch** (a carbohydrate), made in the leaves, which are usually broad and flat to capture sunlight.

- Leaves are green due to the pigment **chlorophyll**, which captures sunlight efficiently.

Starch test on a leaf (Activity 10.2, teacher demonstration)
Boil a leaf in water for 5 minutes to soften it.

Remove colour
Put it in a test tube of alcohol standing in a beaker of boiling water until the leaf is colourless. Never heat alcohol directly: it is flammable.

Add iodine
Spread the leaf and add dilute iodine. **Blue-black = starch present.**

The leaf is decolourised so the colour change with iodine can be seen easily.

▸ Role of sunlight and chlorophyll (Activity 10.3)

Bhaskar took leaves with green and non-green patches from two plants, one kept in sunlight and one in the dark for 36 hours. He sketched the patches and did the iodine test.

Plant	Result of iodine test
Kept in sunlight	Green patches turn blue-black; non-green patches do not (insufficient chlorophyll)
Kept in the dark	No blue-black colour, even on green patches

Conclusion: starch is made only in the green (chlorophyll) parts and only in light. Chlorophyll helps in preparing starch in the presence of sunlight, hence leaves are **'food factories'**. Red, violet or brown leaves have other pigments that hide the green; some of those pigments also help photosynthesis, and the iodine test still shows starch.

▸ 10.2.2 Role of air (Activity 10.4, teacher demonstration)

Destarch the plant
Keep a potted plant in the dark for 2–3 days to use up its stored starch.

Set up
Put caustic soda (sodium hydroxide, which absorbs carbon dioxide) in a bottle and insert half of a destarched leaf through a split cork; the other half stays outside. Handle caustic soda carefully: only the teacher should.

Sunlight
Keep it in sunlight for a few hours, then do the iodine test.

Result
The outside half turns blue-black (starch). The half inside the bottle does not, because carbon dioxide has been absorbed.

IMPORTANT

Carbon dioxide from the air is essential for making starch.

DEFINITION

Photosynthesis is the process by which plants prepare food (glucose) from carbon dioxide and water in the presence of **sunlight and chlorophyll**. The leaf is the main site, but any other green part also photosynthesises.

▸ Release of oxygen (Activity 10.5)

A water plant under an inverted funnel and test tube is kept in sunlight (set-up A) and in the dark (set-up B). In sunlight bubbles collect in the test tube. A lit matchstick brought to the gas burns **intensely**, showing it is rich in **oxygen**. So **oxygen is released during photosynthesis, which needs sunlight.**

▸ 10.2.3 Photosynthesis in a nutshell

> **FORMULA**
> Carbon dioxide + Water → (sunlight, chlorophyll) → Glucose + Oxygen

- Water and minerals come from the soil; carbon dioxide from air; sunlight is the energy source.
- Glucose is an instant source of energy and is later converted into **starch** for storage.
- **Know a scientist:** Rustom Hormusji Dastur (1896–1961), head of Botany at the Royal Institute of Science, Bombay, studied how water, temperature and colour of light affect photosynthesis.

▸ 10.2.4 Gas exchange: stomata (Activity 10.6)

A thin peel from the lower surface of a leaf (rhoeo, money plant, hibiscus etc.) examined under a microscope shows tiny pores called **stomata**. Stomata on leaf surfaces allow the exchange of gases (carbon dioxide in, oxygen out in photosynthesis; the reverse in respiration).

10.3 Transport in Plants

▸ 10.3.1 Transport of water and minerals (Activity 10.7)

Set up
Fill two tumblers one-third with water; add red ink to B. Cut the stems of white-flowered twigs (white balsam, sadabahar) obliquely under water and place one in each.

Next day
The twig in red water has red colour in stem, leaves and flowers.

Cut the stem
A cross-section shows red in thin tubes, the **xylem**.

Water with dissolved minerals taken up by the roots moves upward to all parts through the **xylem**, thin tube-like structures in the roots, stem, branches and leaves.

▸ 10.3.2 Transport of food

Food made in leaves is carried to all parts through another set of tubes, the **phloem**. Food may be stored in seeds and roots.

	Xylem	Phloem
Carries	Water and minerals	Food (made in leaves)
Direction	From roots upwards to leaves and other parts	From leaves to all parts, including storage organs
Evidence	Red ink rises in the stem	Food stored in roots and seeds

10.4 Do Plants Respire?

Set up (Activity 10.8, demonstration)
Place soaked moong seeds on wet cotton in a conical flask with a two-holed cork and tubes; connect to a test tube of lime water. Another test tube of lime water is left as comparison.

Wait 24 hours in the dark
Prevent photosynthesis.

Observe
Lime water connected to the seed flask turns **milky**: germinating seeds released extra carbon dioxide by respiration.

FORMULA
Glucose + Oxygen → Carbon dioxide + Water + Energy (respiration)

- Energy released is used for plant growth and development.
- **All parts of a plant, green or non-green, respire.**

Feature	Photosynthesis	Respiration
Raw materials	Carbon dioxide, water, sunlight, chlorophyll	Glucose, oxygen
Products	Glucose (then starch), oxygen	Carbon dioxide, water, energy
Energy	Light energy is stored in food	Energy is released from food
Occurs	Only in green parts and in light	In all living cells, day and night
Gas exchange	Takes in CO_2, gives out O_2	Takes in O_2, gives out CO_2
Importance	Makes food and oxygen; basis of food chains	Provides energy for life processes

Know a scientist: Kamala Sohonie (1911–1998) earned a Ph.D. from Cambridge for work on plant respiration, worked at Lady Hardinge Medical College, the Nutrition Research Laboratory (Coonoor) and the Royal Institute of Science (Bombay) and developed nutritious *Neera* from coconut palm sap.

Likely exam questions

Likely question	Model answer
What do plants need for growth according to Activity 10.1?	Sunlight and water (besides air and nutrients).
Why are leaves called food factories?	They contain chlorophyll and make food (starch/glucose) by photosynthesis using sunlight.
How do you test a leaf for starch?	Decolourise it in alcohol (water bath), add iodine; blue-black shows starch.
Why is the leaf decolourised before the iodine test?	So that the colour change is easily seen.
What does the leaf with green and non-green patches show?	Starch forms only in green parts and in sunlight, so chlorophyll is needed.
Why is caustic soda used in the experiment on air?	It absorbs carbon dioxide, showing carbon dioxide is needed for photosynthesis.
Define photosynthesis and write its equation.	Food making by plants; carbon dioxide + water gives glucose + oxygen in sunlight with chlorophyll.
How do we know oxygen is released in photosynthesis?	Gas from a sunlit water plant relights/intensifies a matchstick flame.

Likely question	Model answer
What are stomata?	Tiny pores on leaves for gas exchange.
Differentiate xylem and phloem.	Xylem carries water and minerals up; phloem carries food from leaves to all parts.
Do plants respire? How do you show it?	Yes; germinating seeds turn lime water milky because they release carbon dioxide.
Write the equation of respiration.	Glucose + oxygen gives carbon dioxide + water + energy.
What would happen if all photosynthetic organisms disappeared?	No food and no oxygen replenishment; animals and humans could not survive.
Where does potato starch come from?	Made in leaves by photosynthesis and carried by phloem to the potato for storage.

Exploratory projects

- Bottle garden: a sealed bottle with a plant survives because photosynthesis and respiration balance gases.
- Study how photosynthesis, respiration and transport matter in crop production.
- Visit a greenhouse and find how light, water and CO_2 are regulated.
- Design an experiment on whether water transport is quicker in warm or cold conditions.

★ KEY TAKEAWAYS

- Plants need **sunlight, water, carbon dioxide and chlorophyll** to make food.
- **Photosynthesis:** CO_2 + water → glucose + oxygen (sunlight, chlorophyll); glucose is stored as **starch**; starch is detected by **iodine (blue-black)**.
- Leaves are the **food factories**; **stomata** exchange gases.
- **Xylem** carries water and minerals; **phloem** carries food.
- Plants **respire** in all parts: glucose + oxygen → CO_2 + water + energy.
- Photosynthesis and respiration keep a balance of gases in nature.

CHAPTER 10

Light: Shadows and Reflections

Keshav, watching fireflies and moonlight in the Western Ghats, notices that light seems to travel straight and wonders whether the Moon makes its own light. The chapter covers luminous and non-luminous objects, the straight-line travel of light, transparent, translucent and opaque materials, shadow formation, reflection and images in a plane mirror, the pinhole camera, and useful devices like the periscope and the kaleidoscope.

Big Questions

- Does the Moon make its own light?
- How do we know light travels in a straight line?
- How is a shadow formed?
- How is the image in a plane mirror different from a pinhole-camera image?

11.1 Sources of Light

- The **Sun** is the main natural source of light on Earth. Stars, lightning, natural fire and certain animals (fireflies, which use light to communicate) also give their own light.
- The earliest artificial light was **fire**, later made with animal fat, oil, wax and gas; now most lighting is electric.
- **LED lamps** use much less power, are brighter and last longer than older lamps; the Government of India promotes them nationwide. Dispose of or recycle them properly at the end of life.
- Fireflies are decreasing because of light pollution, reduced forest cover and excess tourism.

Luminous objects	Non-luminous objects
Emit their own light: Sun, stars (Pole Star), lightning, fire, lamp, firefly	Do not emit light; seen only by reflecting light: Moon, planets (Mars, Venus), mirror, book, table

IMPORTANT

The **Moon is non-luminous**; moonlight is reflected sunlight. Planets such as Mars and Venus also shine by reflected sunlight.

11.2 Does Light Travel in a Straight Line?

Three matchboxes (Activity 11.1)
Make holes at the same position in the inner trays and line up the three boxes with holes at the same height.

Shine a torch
Place a torch at one end and a cardboard screen at the other. A bright spot appears.

Move one box
Shift one matchbox aside or up and down: the spot disappears.

- **Bent pipe test (Activity 11.2):** a candle flame can be seen through a straight pipe but not through a bent one.

- A laser beam in water made visible with a drop of milk follows a straight path. Never point a laser at anyone's eyes; use a low-power laser only under teacher supervision.
- Light can bend around corners in certain situations, to be learnt later.

> **IMPORTANT**
> **Light travels in a straight line.**

11.3 Transparent, Translucent and Opaque Materials

Type	Light passes	Examples
Transparent	Almost completely	Clear glass, clear water, air
Translucent	Partially	Tracing paper, thin paper, frosted glass, oiled paper
Opaque	Not at all	Cardboard, thick cloth, wood, metal

Activity 11.3: place each material between a torch and a screen; predict first, then observe whether the spot of light appears fully, partly, or not at all.

11.4 Shadow Formation

> **DEFINITION**
> A **shadow** is the dark region formed when an opaque object blocks the straight-line path of light from falling on a screen.

- Opaque objects form **dark** shadows; translucent objects make **lighter** shadows; even some transparent objects can form faint shadows.
- **Three things are needed:** a light source, an opaque object and a screen (wall, floor or ground).

Action (Activity 11.4)	Effect on shadow
Screen removed	No shadow can be seen
Object removed	No shadow
Torch switched off	No shadow
Object moved closer to screen	Shadow becomes smaller and sharper
Object moved closer to torch	Shadow becomes larger and fuzzier
Object tilted	Shape and size of shadow change
Colour of object changed	Colour of shadow stays the same (dark)

- Shape, size and sharpness depend on where the object is relative to the source and the screen.
- A shadow may or may not reveal what the object is.
- With the torch close to the object the shadow is larger; with the torch far, smaller. Two torches give two shadows.
- A bird high in the sky has a tiny, faint shadow on the ground; near the ground its shadow is clear. The length of your shadow is shortest at noon, when the Sun is highest.

- **Shadow play** (shadow puppetry) is a traditional art: Charma Bahuli Natya (Maharashtra), Tholu Bommalata and Keelu Bomme (Andhra Pradesh), Togalu Gombeyaata (Karnataka), Ravana Chhaya (Odisha), Tholpavakoothu (Kerala), Bommalattam (Tamil Nadu).

11.5 Reflection of Light

A shiny plate or plane mirror kept in sunlight can send a spot of light on a wall. It changes the direction of light that falls on it. Tilting it moves the spot.

> DEFINITION
> **Reflection of light** is the change in direction of light when it falls on a shiny surface such as a mirror. A **plane mirror** is a flat mirror.

Activity 11.6: a comb with all but one slit covered by black paper gives a thin beam over white paper. A mirror put in its path changes the beam's direction.

11.6 Images Formed in a Plane Mirror

The **object** (e.g. a pen) is in front of the mirror; its **image** appears to be behind the mirror.

Properties of image in a plane mirror	Meaning
Same size as the object	Image is neither magnified nor reduced
Erect	Upright; top of the object is at top of image
Cannot be obtained on a screen	It appears behind the mirror, so no screen can catch it
Image distance = object distance	Image is as far behind the mirror as the object is in front
Laterally inverted	Left appears right and right appears left

- Raise your left arm and the image raises its right arm: **lateral inversion**.
- 'AMBULANCE' is written laterally inverted on the front of the ambulance so that it reads correctly in the rear-view mirrors of vehicles ahead.
- Letters like T and O look identical in a mirror; others do not.
- **Aranmula Kannadi** in Kerala is a metal-surface mirror made for centuries.

11.7 Pinhole Camera

> DEFINITION
> A **pinhole camera** is a device in which light from an object passes through a tiny hole and forms an image on a screen.

- A lighted candle in front of a cardboard with a pinhole forms an **inverted (upside-down)** image of the flame on a screen.
- **Sliding pinhole camera (Activity 11.10):** two boxes fit one into the other. The larger has a pinhole in the middle of one face; the smaller has a 5–6 cm square on its opposite face covered with tracing paper as a screen. Look at a bright distant object through the open side, covering your head with dark cloth, and slide the boxes until a sharp image forms. The image is coloured and inverted.
- Mirror image: laterally inverted but not upside down. Pinhole image: upside down.

Feature	Plane mirror image	Pinhole camera image
Upright/inverted	Erect	Inverted
Lateral inversion	Yes	Yes (also upside down)
Size	Same as object	Depends on distances
Obtained on screen?	No (virtual)	Yes

11.8 Making Useful Items

Item	Construction	Use
Periscope	Two plane mirrors fixed in a Z-shaped (or tube) box, one at each end, facing each other at an angle; light reflects twice	See objects not directly visible; used in submarines, tanks, bunkers, or to look over tall friends
Kaleidoscope	Three equal mirror strips joined in a triangle inside a tube; coloured bangle bits or beads on a transparent sheet at one end; tracing paper cover	Multiple reflections give changing patterns every time it is turned; used by designers for pattern ideas

NOTE

Fascinating fact: light from the Sun takes about **8 minutes 20 seconds** to reach the Earth.

Likely exam questions

Likely question	Model answer
Differentiate luminous and non-luminous objects.	Luminous objects emit their own light (Sun, stars); non-luminous do not (Moon, planets, mirror).
Why does the Moon shine?	It reflects sunlight.
How would you show that light travels in a straight line?	Three-matchbox holes in line give a spot only when aligned; or flame visible only through a straight pipe.
Classify transparent, translucent and opaque with examples.	Transparent (glass), translucent (tracing paper), opaque (cardboard).
What is a shadow? What is needed to observe one?	Dark region where an opaque object blocks light; source, opaque object and screen.
Does colour of the object change the shadow colour?	No.
What happens to the shadow when the object is moved closer to the torch?	Becomes larger and less sharp.
Define reflection of light.	Change in direction of light by a shiny surface like a mirror.
List properties of an image in a plane mirror.	Same size, erect, laterally inverted, same distance behind the mirror, cannot be got on a screen.
What is lateral inversion? Give an example.	Left appears right and right left; the word AMBULANCE written reversed.

Likely question	Model answer
What is a pinhole camera and what image does it form?	Light passing through a tiny hole forms an inverted image on a screen.
How does a periscope work?	Two plane mirrors reflect light twice so one can see over an obstacle.
Why can we see many patterns in a kaleidoscope?	Reflections in three mirrors produce multiple images that rearrange when turned.
Why do we not see the shadow of a high-flying bird?	Being far from the ground, its shadow is very faint and diffuse.

Exploratory projects

- Find why fireflies are no longer seen in your area and write a story about it.
- Use coloured transparent paper over a torch and report the shadow colour.
- Place two mirrors at an angle or parallel and count the images.
- Will a small mirror form an image of a large tree? Predict, then test.

★ KEY TAKEAWAYS

- Luminous objects emit light; the **Moon** and planets only reflect sunlight.
- **Light travels in a straight line.**
- Materials are **transparent, translucent or opaque**.
- A **shadow** needs a source, an opaque object and a screen; its size and sharpness depend on positions.
- **Reflection** changes the direction of light at a mirror.
- A plane mirror image is **same size, erect, laterally inverted**, formed as far behind as the object is in front, and cannot be caught on a screen.
- A **pinhole camera** forms an **inverted** image; **periscope** and **kaleidoscope** use mirrors.

CHAPTER 11

Earth, Moon, and the Sun

Rashmika in Kanniyakumari notices coconut-tree shadows are long in the morning and short at noon, and wonders: does the Sun move, or does the Earth? The chapter explains the Earth's rotation (day and night, apparent motion of the Sun, Moon and stars), its revolution (changing night sky, seasons due to tilt), and how the Sun, Moon and Earth line up to cause solar and lunar eclipses.

Big Questions

- Why does the Sun rise in the East?
- What causes day and night?
- Why do seasons occur?
- How can a small Moon hide the huge Sun?

12.1 Rotation of the Earth

The Sun appears to rise in the East, cross the sky and set in the West. **Merry-go-round activity (12.1):** when you turn anti-clockwise on a merry-go-round, objects around you appear to turn the opposite way (clockwise). In the same way, the Sun *appears* to move because we view it from a turning Earth.

DEFINITION

Rotation is the motion in which all parts of an object move in circles around an imaginary line through it, the **axis of rotation**. Examples: a spinning top, fan or ball.

- The Earth spins on its axis, which passes through the geographic **North and South Poles**.
- One rotation takes about **24 hours**.
- Seen from above the North Pole, the Earth turns **anti-clockwise, i.e. from West to East**.

Globe activity (12.2)
Mark your location with a sticker on a globe and turn it anti-clockwise as seen from above the North Pole until the sticker returns to its place.

Add a torch
Shine a torch from about 1.5 m. Half the globe is lit (day) and half dark (night).

Check direction
Light falls on the eastern part of India first only when the globe turns West to East.

Observe
As your location turns into light there is sunrise; turning into darkness is sunset.

IMPORTANT

Day and night are caused by the Earth's rotation: the side facing the Sun has day, the other side night.

▸ What an observer on the rotating Earth sees

Morning: the Sun appears on the East side. Noon: Sun overhead. Evening: it disappears in the West and stars appear. Hence the Sun, and also the **Moon**, appear to rise in the East and set in the West.

▸ Apparent motion of stars (Activity 12.3)

- Draw the position of the Big Dipper (Saptarishi) relative to the Pole Star every two hours on the same night. It appears to move around the **Pole Star (Dhruva Tara)**.
- The Earth's axis points almost at the Pole Star, so it looks nearly **stationary** while all stars seem to circle it. Long-exposure photos show star trails as circular arcs.
- **Foucault pendulum:** Leon Foucault (mid-19th century) used a long pendulum to demonstrate the Earth's rotation. A 22 m Foucault pendulum hangs in the Constitution Hall of the new Parliament building in New Delhi.
- **Aryabhata** (*Aryabhatiya*, about 5th century CE) explained the stars' apparent motion with a boat analogy and gave the rotation period as about 23 h 56 min 4.1 s, very close to the modern value.

12.2 Revolution of the Earth

DEFINITION

Revolution is the motion of an object around another object. The path it follows is its **orbit**. The Earth's orbit around the Sun is nearly circular when seen from above. The Earth completes one revolution in about **365 days 6 hours** (about one year).

	Rotation	Revolution
Motion	Spinning on own axis	Going around the Sun
Time taken	About 24 hours	About 365¼ days
Causes	Day and night; apparent motion of Sun, Moon and stars	Changing night sky through the year; (with tilt) seasons

▸ 12.2.1 Changing view of the night sky

Because the Earth moves around the Sun, the direction we look at in the night sky after sunset changes through the year, so **different stars and constellations are seen in different months**. The Bhil and Pawara people of the Tapi Valley used star patterns as markers for the arrival of monsoon.

▸ 12.2.2 Seasons on the Earth

The Earth's axis is **tilted** relative to its orbit and keeps the same tilt (pointing the same way) throughout the year. The **tilt and the spherical shape** of the Earth cause seasons.

	June (Northern Hemisphere)	December (Northern Hemisphere)
Tilt	Northern Hemisphere tilted **towards** the Sun	Northern Hemisphere tilted **away** from the Sun
Sunrays	Same rays fall on a smaller area, so **more intense**	Spread over a larger area, so **less intense**
Day length	More than 12 hours	Less than 12 hours
Season	**Summer**	**Winter**
Southern Hemisphere	Winter	Summer
North Pole	24 hours of sunlight	24 hours of darkness

IMPORTANT

Wrong reasons: seasons are *not* due to the Earth being closer to the Sun when a hemisphere tilts towards it, nor to the oval orbit. The distance difference is tiny; in fact the Earth is closest to the Sun in **January**.

Date (Northern Hemisphere)	Name	Day length
About 21 June	Summer solstice	Longest day
About 22 December	Winter solstice	Shortest day, longest night
About 21 March	Spring equinox	12 hours
About 23 September	Autumn equinox	12 hours

- At the **North Pole** the Sun rises on 21 March and stays up about six months; it sets about 22 September. The South Pole is opposite: six months of sunshine then six months of darkness.
- At the **equator** there are always 12 hours of daylight and 12 hours of darkness, and the Sun's intensity hardly varies, so **southern India** feels less marked seasons.
- Local geography and nearness to seas also influence seasons.
- If the Earth's axis were not tilted there would be **no seasons** as we know them.

12.3 Eclipses

Mercury and Venus, though between the Earth and Sun, look too small to block the Sun. When Venus passes between the Earth and Sun it appears as a tiny dot (a rare **Transit of Venus**). The **Moon**, a natural satellite of the Earth that revolves around it, can block the Sun.

▸ 12.3.1 Solar eclipse

DEFINITION

A **solar eclipse** occurs when the Moon comes between the Sun and the Earth and blocks sunlight from reaching a part of the Earth.

- **Apparent size** depends on actual size and distance. A thumb at arm's length can hide a friend's head 5 m away (Activity 12.4). The Moon is much smaller than the Sun but much **closer**, so both look about the same size in the sky.
- In the Moon's shadow (a small area on Earth) the Sun is wholly hidden: **total solar eclipse**, when the sky darkens for a few minutes. Where the Moon covers only part of the Sun the observer sees a **partial solar eclipse**.
- The Moon's shadow moves across the Earth because of the Earth's rotation and the Moon's motion, so totality lasts only a few minutes. A 'diamond ring' effect appears just as the Moon moves away.

IMPORTANT

Safe viewing: never look at the Sun directly during an eclipse; it can cause blindness. Sunglasses, binoculars and telescopes are NOT safe. Use solar-eclipse goggles, project the Sun's image with a mirror (teacher-supervised; never reflect light into anyone's eyes), or attend planetarium or astronomy-club viewing events.

- Earlier people feared eclipses and held superstitions (not eating, cooking or going out). Now we know the cause, we need not fear them. Eclipses give scientists a chance to study the Sun.

- An eclipse is called **grahan** in Sanskrit and many Indian languages. The *Surya Siddhanta* gives calculations to predict eclipses.

▸ 12.3.2 Lunar eclipse

> **DEFINITION**
> A **lunar eclipse** occurs when the Earth comes between the Sun and the Moon and its shadow falls on the Moon, blocking sunlight from reaching it.

	Solar eclipse	Lunar eclipse
Order of bodies	Sun – Moon – Earth	Sun – Earth – Moon
What blocks light	Moon	Earth
Shadow falls on	A small part of the Earth	The Moon
Types	Total, partial	Total (Moon appears dark red), partial
Visible from	Only a small region of Earth	Anywhere on the night side of Earth
Safe to see with naked eye?	**No**	Yes
Duration	A few minutes (total)	Much longer

Useful facts: the free Stellarium app tells about upcoming eclipses. **Kodaikanal Solar Observatory** (Palani hills, est. 1899, run by the Indian Institute of Astrophysics, Bengaluru) has Sun data for 100+ years. **M. K. Vainu Bappu**, father of modern Indian astronomy, set up telescopes at Manora Peak (Nainital) and Kavalur (the observatory is named after him), discovered a comet and studied eclipses.

Likely exam questions

Likely question	Model answer
Why does the Sun appear to rise in the East and set in the West?	The Earth rotates from West to East.
Define rotation and axis of rotation.	Rotation: all parts move in circles about a line through the object; that line is the axis.
What causes day and night?	The Earth's rotation on its axis in about 24 hours.
Why does the Pole Star appear stationary?	The Earth's axis points almost at it.
Define revolution and orbit. How long is one revolution of the Earth?	Motion around another object; the path is the orbit; about 365 days 6 hours.
Why are different stars seen in different months?	The Earth revolves around the Sun, so we face different directions of space after sunset.
What causes seasons?	The tilt of the Earth's axis with its spherical shape, causing different sunray intensity and day lengths.
Why is it summer in the Northern Hemisphere in June?	It is tilted towards the Sun, getting more intense light for more than 12 hours.
Is the Earth closer to the Sun in summer?	No; the Earth is closest in January.

Likely question	Model answer
Why are seasons less marked in southern India?	Near the equator day length and sun intensity change little.
Define solstice and equinox.	Solstice: longest (21 June) or shortest (22 December) day; equinox: about 21 March and 23 September with equal day and night.
How can the small Moon cover the large Sun?	It is much closer, so apparent sizes are similar.
Differentiate a solar and a lunar eclipse.	Solar: Moon between Sun and Earth; lunar: Earth between Sun and Moon.
Why is it unsafe to view a solar eclipse directly?	Sunlight can damage the eyes and cause blindness, even with sunglasses.
Why is a lunar eclipse visible from a larger area than a total solar eclipse?	The Earth's shadow covers the whole Moon, seen by all on the night side; the Moon's shadow on Earth is small.
Should the Indian cricket team pack winter or summer clothes for Australia in December?	Summer clothes; December is summer in the Southern Hemisphere.

Exploratory projects

- Repeat the globe activity with an electric lamp and a tilted globe at four positions on a circle to see how much of each hemisphere is lit and how day length changes.
- Draw circles of radius 14.7 cm and 15.2 cm (1 cm = 10 million km) to see how small the Earth's closest and farthest Sun distances differ.
- Find the tilt of Uranus and its seasons; write a short article.
- Look at the textbook covers for hidden science clues (more than 25).

★ KEY TAKEAWAYS

- The Earth **rotates** West to East in about **24 hours**, causing **day and night** and the apparent motion of the Sun, Moon and stars.
- The Earth **revolves** around the Sun in about **365¼ days**; the night sky changes through the year.
- **Seasons** result from the **tilt** of the axis and the spherical shape of the Earth, not from changes in distance.
- Solstices (about 21 June, 22 December) give longest and shortest days; equinoxes (about 21 March, 23 September) give equal day and night.
- **Solar eclipse:** Moon between Sun and Earth; **lunar eclipse:** Earth between Sun and Moon.
- Never watch a solar eclipse directly; lunar eclipses are safe to view.
- The Moon looks as big as the Sun because it is much closer.